RED HAT ENTERPRISE LINUX

Understanding & Creating Swap Space

A practical guide to virtual memory, swap files, swap partitions, and disk-based swap management in RHEL



10 SLIDES · SYSTEM ADMINISTRATION

What Is Swap Space?

A reserved area of disk that extends your system's usable memory

Definition

Swap space is disk storage used by the Linux kernel as an overflow for RAM.

When physical memory (RAM) fills up, inactive pages are moved to swap, freeing RAM for active processes.

It is slower than RAM but prevents applications from crashing due to out-of-memory errors.

RAM	SWAP
Fast	Slower
Volatile — cleared on reboot	Persistent on disk until removed

Where it lives

- A dedicated swap partition, or
- A swap file on an existing filesystem

Why Swap Space Is Needed

Four reasons swap remains a standard part of RHEL system design

Prevents OOM crashes

Gives the kernel room to move idle pages out of RAM instead of killing processes when memory runs low.

Supports hibernation

Suspend-to-disk writes the entire RAM image into swap so the system can resume later.

Extends effective memory

Lets a system handle memory spikes beyond installed RAM, useful on smaller VMs and servers.

Improves stability under load

The kernel can proactively swap out rarely used pages, keeping more RAM free for disk cache.

Two Ways to Configure Swap

RHEL supports swap on a dedicated partition or inside a regular file

Swap Partition

A dedicated disk partition (type Linux swap) used exclusively for swap.

PROS

Slightly faster; simple, predictable performance; standard on physical servers.

CONS

Fixed size; resizing requires repartitioning or LVM.

Swap File

A regular file on an existing filesystem, formatted and enabled as swap.

PROS

Easy to create, resize, or remove without touching partitions. Great for VMs and cloud instances.

CONS

Marginally slower on some filesystems (not on XFS/ext4 with proper setup).

Check Current Swap Status

Always confirm what's already configured before adding more swap

```
# View swap usage summary
$ free -h

# List active swap devices
$ swapon --show

# Detailed memory + swap info
$ cat /proc/swaps

# Check block devices & sizes
$ lsblk
```

What to look for

Total vs. Used

"free -h" shows how much swap exists and how much is in use right now.

Active devices

"swapon --show" lists every partition or file currently acting as swap, with type and priority.

Free disk space

Confirm there's enough free space on the target filesystem before creating a swap file.

Existing partitions

"lsblk" reveals unpartitioned or unused space that could become a swap partition.

Create a Swap File — Step by Step

The most flexible way to add swap on RHEL

1 Allocate the file

```
sudo fallocate -l 2G /swapfile
```

Reserves 2 GB. If fallocate isn't supported by the filesystem, use dd instead.

2 Secure the permissions

```
sudo chmod 600 /swapfile
```

Only root should be able to read or write the swap file.

3 Format as swap

```
sudo mkswap /swapfile
```

Writes a swap signature so the kernel recognizes the file.

4 Enable it

```
sudo swapon /swapfile
```

Activates the swap file immediately for the current session.

Create a Swap Partition

Use a dedicated partition when you need consistent, predictable performance

```
# 1. Create the partition (interactive)
$ sudo fdisk /dev/sdb
  Command (m for help): n     -> new partition
  Command (m for help): t     -> change type
  Hex code: 82                -> Linux swap
  Command (m for help): w     -> write & exit

# 2. Format the partition as swap
$ sudo mkswap /dev/sdb1

# 3. Activate it
$ sudo swapon /dev/sdb1
```

Good to know

Use a blank or unused disk/partition — this process is destructive to existing data.

On GPT disks, parted can be used instead of fdisk with type code 8200.

LVM logical volumes can also be formatted as swap the same way with mkswap.

Make Swap Permanent

swapon activates swap only until reboot — /etc/fstab makes it stick

```
# Back up fstab first
$ sudo cp /etc/fstab /etc/fstab.bak

# Append the swap entry
$ echo '/swapfile none swap sw 0 0' \
  | sudo tee -a /etc/fstab
```

For a partition, use its UUID instead of a path:

```
UUID=xxxxxxxx-xxxx  none  swap  sw  0 0
```

Find it with: blkid /dev/sdb1

fstab fields, explained

`Device`	Path or UUID of the swap file/partition
`none`	Swap has no mount point
`swap`	Filesystem type
`sw`	Mount option for swap
`0 0`	Dump and fsck order — always 0 for swap

Test it without rebooting:

```
sudo mount -a && sudo swapon -a
```

Managing & Tuning Swap

Common day-two operations once swap is up and running

Disable a swap file/partition

```
sudo swapoff /swapfile
```

Deactivates swap; do this before deleting it.

Remove a swap file

```
sudo rm /swapfile
```

Also delete its line from /etc/fstab.

Check swappiness

```
cat /proc/sys/vm/swappiness
```

0-100 scale; controls how aggressively the kernel swaps (default 60).

Adjust swappiness

```
sudo sysctl vm.swappiness=10
```

Lower values favor RAM; set permanently in /etc/sysctl.conf.

Best Practices & Recap

Sizing guidance and the key takeaways from this deck

Recommended Swap Sizing (RHEL guidance)

≤ 2 GB RAM	**2x RAM**
2-8 GB RAM	**Equal to RAM**
8-64 GB RAM	**0.5x RAM**
> 64 GB RAM	**4-8 GB (or per workload)**

Hibernation systems need swap ≥ total RAM, regardless of the table above.

Key Takeaways

- ✓ Swap extends RAM using disk — slower, but prevents OOM crashes
- ✓ Choose a swap file for flexibility, a partition for raw performance
- ✓ mkswap formats, swapon activates, /etc/fstab makes it permanent
- ✓ Verify with free -h and swapon --show after every change
- ✓ Size swap to workload — more isn't always better